

Thomas Zhong

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EDUCATION

University of Maryland | GPA: 3.54

B.S. Computer Science; Specialization in Machine Learning

Aug. 2023 - Present

Selected Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Data Science, Algorithms, Computational Game Theory, Linear Algebra, Calculus III, Applied Probability and Statistics, AWS

PROFESSIONAL EXPERIENCE

Scale AI

Generative AI Intern

May 2025 - Sep 2025

- Facilitated Reinforcement Learning from Human Feedback (**RLHF**) for generative AI systems.
- Conducted systematic **prompt engineering** to maximize misalignment and hallucinations.
- Solved expert-level SWE GitHub issues and competitive programming challenges in **Python**.

Glenbrook Trading

Quantitative Researcher

Mar. 2025 - Sep 2025

- Analyzed order book flow and market inefficiencies in **Python** through **quantitative modeling**.
- Wrangled **hundreds of millions** of raw order data to support medium frequency trading strategies.
- Investigated and engineered novel features for **Kalman filters** and algorithmic trading execution.

DREAM Malware Lab

Machine Learning Research Intern

Jun. 2022 - Sep. 2024

- Transformed the **8 terabyte** SOREL-20M malware dataset into Burrows-Wheeler Markov Distance (BWMD) vectors, increasing the efficiency and effectiveness of feature extraction.
- Trained a neural network to classify malware files into 11 categories with **92% accuracy**.
- Engineered a CNN with **positional encoding** and **transformers** to interpolate frames within videos.

EXTRACURRICULARS

Maryland AI Safety Initiative

Founder and President

- Leading UMD's first AI safety initiative (60+ members) preparing students for technical or policy AI safety careers through reading groups and mentored research.

Kairos Pathfinder Fellowship (AI Safety Organizers)

- Received extensive training and mentorship for organizing an AI safety group.

Algoverse AI Safety Research Fellow

- Co-authoring a research paper on mechanistic interpretation approaches to prevent subliminal learning.
- Identified behavior-associated latent vectors with **principle component analysis**, **mean activation differences**, and **sparse autoencoders**.
- Steered model behavior with concept-ablation **fine tuning** (currently insignificant at 95% confidence)

Internal Revenue Service: Volunteer Income Tax Assistance (VITA)

- Helping people from marginalized groups do their taxes.

SKILLS/TECHNOLOGIES

Expert: Python (Pandas, Numpy, Pytorch), Generative AI and LLMs, other Machine Learning models

Proficient: Java, C, SQL, R, MATLAB, Jupyter Notebooks, Git, AWS, JSON, Pickle